

Marking Scheme

2026 · Paper Paper 1 · 105 marks

General Rules

- M / A / PROOF mark types
- **f.t.** : M marks always follow through. A marks follow through ONLY when isFollowThrough = true.
- Deduction cap : $A(1)=1 \cdot A(2)=1 \cdot B=0$ (u, pp)
- No deduction in any step scoring 0.
- A correct answer merits all marks unless a method is specified.
- Different notation is NOT penalized. · Benefit of doubt
- All fractional answers must be simplified.

Section A(1) (35 marks)

1. Laws of Indices [B-]

(a) (3 marks)

M1 $(p^{-1}q)^3 = p^{-3}q^3$
for expanding the power / 展開次方

M1 $\frac{p^7q^{-2}}{p^{-3}q^3} = p^{7-(-3)}q^{-2-3} = p^{10}q^{-5}$ **f.t.**
subtract indices / 相除時指數相減

A1 $= \frac{p^{10}}{q^5}$
positive indices required / 必須以正指數表示

2. Change of Subject [B-]

(a) (3 marks)

M1 $3h + 2 = k(2h - 5)$
cross multiplication / 交叉相乘

M1 $3h - 2kh = -5k - 2 \Rightarrow h(3 - 2k) = -(5k + 2)$ **f.t.**
group h terms / 集合含 h 的項

A1 $h = \frac{5k+2}{2k-3}$
or equivalent

3. Estimation and Error [B-]

(a) (1 mark)

A1 $340 - \frac{20}{2} = 330 \text{ g}$
u-1 if unit g missing / 漏寫單位 g 扣 1 分

(b) (2 marks)

M1 **Greatest possible total = $5 \times 350 = 1750 \text{ g}$** **f.t.**
use upper limit 350 g / 使用上限 350 g

A1 **$1740 < 1750 \Rightarrow$ it IS possible** **f.t.**
1A f.t. — conclusion required / 結論句必須寫出

4. Factorization [B]

(a) (2 marks)

M1 $16x^2 - 24xy + 9y^2 = (4x)^2 - 2(4x)(3y) + (3y)^2$
recognize perfect square / 辨認完全平方式

A1 $= (4x - 3y)^2$

(b) (2 marks)

M1 $= (4x - 3y)^2 + 2(4x - 3y)$ f.t.
use result of (a) / 引用 (a)

A1 $= (4x - 3y)(4x - 3y + 2)$
pp-1 if not fully factorized / 未完全分解扣 1 分

5. Percentages [B]

(a) (4 marks)

M1 Let C be the cost. $30\% \times C = 240$
set up profit equation / 建立利潤方程

A1 $C = 800$

M1 $\text{Selling price} = 800 + 240 = 1040$; $MP \times (1 - 20\%) = 1040$ f.t.
SP = MP x 80% / 售價 = 標價 x 80%

A1 $\text{Marked price} = 1300$
u-1 if missing / 漏寫扣 1 分

6. Linear Inequalities [B]

(a) (3 marks)

M1 $-2(x + 1) \geq x - 11 \Rightarrow -3x \geq -9 \Rightarrow x \leq 3$
reverse sign dividing by -3 / 除以負數反轉不等號

M1 $4x + 1 > x - 5 \Rightarrow 3x > -6 \Rightarrow x > -2$

A1 **Taking the intersection:** $-2 < x \leq 3$ f.t.
'and' -> intersection / 「且」取交集

(b) (1 mark)

A1 **3** f.t.
1A f.t. from (a)

7. Transformation in the Rectangular Coordinate Plane [B+]

(a) (2 marks)

A1 $A' = (-7, -3)$
reflection about y-axis: $(x, y) \rightarrow (-x, y)$

A1 $B' = (5, 2)$
clockwise 90 degrees: $(x, y) \rightarrow (y, -x)$

(b) (2 marks)

M1 $\text{slope} = \frac{2 - (-3)}{5 - (-7)}$ f.t.
1M f.t. from (a)

A1 $= \frac{5}{12}$

8. Congruent Triangles [B+]

(a) (2 marks)

PROOF $AB = AD$ (given) ; $\angle BAC = \angle DAC$ (given) ; $AC = AC$ (common)

Case 1 with reasons = 2; Case 2 without reasons = 1

PROOF $\therefore \triangle ABC \cong \triangle ADC$ (SAS)

reason SAS required for 2nd mark / 須寫理由「邊、角、邊」

(b) (3 marks)

M1 $\angle ADC = \angle ABC = 104^\circ$ (corr. $\angle$ s, $\cong \triangle$ s) **f.t.**

use result of (a) / 引用 (a)

M1 $\angle BAD + \angle ABC + \angle BCD + \angle ADC = 360^\circ$

angle sum of quadrilateral / 四邊形內角和

A1 $\angle BAD = 360^\circ - 104^\circ - 104^\circ - 88^\circ = 64^\circ$

u-1 if degree symbol missing / 漏寫「度」扣1分

9. Statistical Tables and Probability [B+]

(a) (1 mark)

A1 21 – 25 minutes

(b) (2 marks)

M1 $\text{Mean} = \frac{5(13)+11(18)+13(23)+7(28)+4(33)}{40} = \frac{890}{40}$

use class marks / 使用組中點

A1 = 22.25 minutes

u-1 if unit missing / 漏寫單位扣1分

(c) (2 marks)

M1 $P = \frac{5+11}{40}$

A1 $= \frac{2}{5}$

simplify fraction / 分數須化至最簡

Section A(2) (35 marks)

10. Variation [A-]

(a) (3 marks)

M1 $f(x) = a + bx^2, b \neq 0$

correct form / 部分變分形式

M1 $\begin{cases} a + b = 9 \\ a + 16b = 54 \end{cases} \Rightarrow 15b = 45$ **f.t.**

form simultaneous equations / 建立聯立方程

A1 $b = 3, a = 6 \Rightarrow f(x) = 6 + 3x^2$

both correct / 兩者皆正確

(b) (1 mark)

A1 6 **f.t.**

1A f.t. from (a)

(c) (3 marks)

M1 $6 + 3x^2 = k \Rightarrow x^2 = \frac{k-6}{3}$ **f.t.**

M1 Two distinct real roots $\Leftrightarrow \frac{k-6}{3} > 0$ **f.t.** **alt: q10c_alt**

or use discriminant / 或用判別式

A1 $k > 6$

11. Measures of Central Tendency and Dispersion [A-]

(a) (3 marks)

M1 **Median = the 6th datum = $20 + a = 27 \Rightarrow a = 7$**

median = 6th of 11 data / 中位數為 11 個數據的第 6 個

M1 **$Q_1 = 19$, $Q_3 = 30 + b$, $Q_3 - Q_1 = 16 \Rightarrow b = 5$ **ft.****

$Q_1 = 3rd$ datum, $Q_3 = 9th$ datum / Q_1 為第 3 個、 Q_3 為第 9 個

A1 **$a = 7$, $b = 5$**

both correct / 兩者皆正確

(b) (2 marks)

A1 **Mean = $\frac{305}{11} \approx 27.7$ **ft.****

r.t. 27.7

A1 **Standard deviation ≈ 9.33 **ft.****

1A r.t. 9.33

r.t. 9.33

(c) (2 marks)

M1 **New median = $\frac{5th + 6th \text{ of the remaining 10 data}}{2} = \frac{25+27}{2} = 26$ **ft.****

A1 **The median decreases by $27 - 26 = 1$ **ft.****

1A f.t. — direction must be stated / 須說明增減方向

12. Remainder Theorem and Factor Theorem [A]

(a) (3 marks)

M1 **$p(x) = (x^2 - x + 3)(2x + t) + (-8x - 3)$**

division algorithm, quotient $2x + t$ / 除法算式, 商為 $2x + t$

M1 **Constant: $3t - 3 = -12 \Rightarrow t = -3$ **ft.****

compare coefficients / 比較係數

A1 **$a = t - 2 = -5$, $b = 1$**

both correct / 兩者皆正確

(b) (2 marks)

M1 **$p(3) = 2(27) - 5(9) + 3 - 12$ **ft.****

factor theorem / 因式定理

A1 **$p(3) = 54 - 45 + 3 - 12 = 0 \Rightarrow x - 3$ IS a factor **ft.****

1A f.t. — conclusion required / 結論句必須寫出

(c) (2 marks)

M1 **$p(x) = (x - 3)(2x^2 + x + 4)$; $\Delta = 1^2 - 4(2)(4) = -31 < 0$ **ft.****

1M for factorization OR discriminant / 因式分解或判別式

A1 **$\Delta < 0 \Rightarrow 2x^2 + x + 4 = 0$ has no real root; only ONE real root $x = 3 \Rightarrow$ DISAGREE**

ft.

1A f.t. — conclusion required / 結論句必須寫出

13. Volumes and Surface Areas of Similar Solids [A]

(a) (3 marks)

M1 **Volume of larger cone** $= \frac{1}{3}\pi(18)^2(8) = 864\pi \text{ cm}^3$
cone volume formula / 圓錐體積公式

M1 $\frac{V_{\text{small}}}{864\pi} = \frac{8}{27}$ **ft.**
use volume ratio / 運用體積比

A1 **$V_{\text{small}} = 256\pi \text{ cm}^3$**
u-1 if cm^3 missing / 漏寫 cm^3 扣 1 分

(b) (2 marks)

M1 **$\pi(4)^2h = 864\pi + 256\pi$** **ft.**
conservation of volume / 體積守恆

A1 **$h = \frac{1120}{16} = 70 \text{ cm}$**
u-1 if cm missing / 漏寫 cm 扣 1 分

(c) (2 marks)

M1 **Curved SA of cylinder** $= 2\pi(4)(70) = 560\pi$; **slant height of larger cone** $= \sqrt{18^2 + 8^2} = \sqrt{388}$ **ft.**
1M f.t. from (b)

A1 **Curved SA of larger cone** $= \pi(18)\sqrt{388} \approx 354\pi$; **$560\pi > 354\pi \Rightarrow \text{AGREE}$** **ft.**
1A f.t. — conclusion required / 結論句必須寫出

14. Equations of Circles [A]

(a) (2 marks)

A1 **$G = (10, 8)$**

A1 **$r = \sqrt{10^2 + 8^2 - 139} = \sqrt{25} = 5$**

(b) (1 mark)

A1 **$GH = \sqrt{(22 - 10)^2 + (8 - 8)^2} = 12$** **ft.**

(c)(i) (1 mark)

A1 **$GP \perp GH$**
accept 'GP perpendicular to GH' / 亦接受「GP 垂直於 GH」

(c)(ii) (3 marks)

M1 **$GP = r = 5$, $GH = 12$, $\angle HGP = 90^\circ$** **ft.**

M1 **$HP = \sqrt{5^2 + 12^2} = 13$** **ft.**
Pythagoras / 畢氏定理

A1 **Perimeter** $= 5 + 12 + 13 = 30$

Section B (35 marks)

15. Permutations, Combinations and Probability [A]

(a) (3 marks)

M1 $\text{Total} = C_4^{15} = 1365$

M1 $\text{Favourable} = C_2^8 \times C_2^7 = 28 \times 21 = 588$ **f.t.**

A1 $P = \frac{588}{1365} = \frac{28}{65}$
simplify fraction / 分數須化至最簡

(b) (3 marks)

M1 $P = 1 - P(\text{no red}) - P(\text{no yellow}) + P(\text{no red and no yellow})$
inclusion-exclusion / 容斥原理

M1 $= \frac{1365 - C_4^7 - C_4^{12} + C_4^4}{1365} = \frac{1365 - 35 - 495 + 1}{1365}$ **f.t.**

A1 $P = \frac{836}{1365}$
1A r.t. 0.612
r.t. 0.612

16. Graphs of Quadratic Functions [A]

(a) (2 marks)

M1 $f(x) = 2(x - 2k)^2 - 8k^2 + 10k^2 + 8k + 5$
completing the square / 配方法

A1 $\text{Vertex} = (2k, 2k^2 + 8k + 5)$

(b) (2 marks)

M1 $2k^2 + 8k + 5 = 2(k + 2)^2 - 3$ **f.t.**
complete the square again / 再次配方

A1 $\text{Least value} = -3 \text{ when } k = -2$
both correct / 兩者皆正確

(c) (2 marks)

M1 Put $x = 2k$ into Γ : $y = \frac{1}{2}(2k)^2 + 4(2k) + 5 = 2k^2 + 8k + 5$ **f.t.**
1M f.t. from (a)

A1 $\text{Equals the } y\text{-coordinate of the vertex} \Rightarrow \text{AGREE}$ **f.t.**
1A f.t. — conclusion required / 結論句必須寫出

17. Arithmetic and Geometric Sequences [A]

(a) (2 marks)

M1 $d = \frac{44-19}{8-3} = 5$
find common difference / 求公差

A1 $A(n) = 19 + (n - 3)(5) = 5n + 4$

(b) (1 mark)

A1 $S(45) = \frac{45}{2}[9 + 229] = 5355$ **f.t.**

(c) (3 marks)

M1 $G(1) = A(2) = 14 \Rightarrow \sum_{i=1}^n G(i) = 14(2^n - 1)$ **f.t.**
sum of geometric sequence / 等比求和

M1 $14(2^n - 1) > 5355 \Rightarrow 2^n > 383.5 \Rightarrow n > \log_2 383.5 \approx 8.58$ **f.t.**
accept trial with $n = 8, 9$ / 亦接受試值

A1 $\text{Least } n = 9 \quad (n = 9 : 7154 > 5355 ; n = 8 : 3570 < 5355)$

18. Trigonometry in 3-Dimensional Problems [A+]

(a) (2 marks)

M1 $OM = 5$; $VM^2 = VO^2 + OM^2 = 12^2 + 5^2$
Pythagoras in triangle VOM / 直角三角形 VOM 用畢氏定理

A1 $VM = \sqrt{169} = 13 \text{ cm}$

(b) (2 marks)

M1 Required angle = $\angle VMO$; $\tan \angle VMO = \frac{12}{5}$ **f.t.**
1M identify correct angle / 正確辨認二面角

A1 $\angle VMO \approx 67.4^\circ$
*1A r.t. 67.4 degrees
 r.t. 67.4*

(c) (4 marks)

M1 $VM \perp BC$ and $VN \perp AD$; $BC \parallel AD \Rightarrow$ required angle = $\angle MVN$
1M identify angle between planes / 辨認兩平面夾角

M1 $VM = VN = 13 \text{ cm}$, $MN = 10 \text{ cm}$ **f.t.**

M1 $\cos \angle MVN = \frac{13^2 + 13^2 - 10^2}{2(13)(13)} = \frac{238}{338}$ **f.t.**
cosine formula / 餘弦公式

A1 $\angle MVN \approx 45.2^\circ > 45^\circ \Rightarrow$ **AGREE** **f.t.**
*1A f.t. r.t. 45.2 degrees — conclusion required
 r.t. 45.2*

19. Equations of Circles and Tangents [A+]

(a) (2 marks)

A1 $G = (3, 4)$

A1 $r = \sqrt{3^2 + 4^2 - 0} = 5$

(b) (3 marks)

M1 Distance from G to $L = \frac{|3(3)+4(4)-50|}{\sqrt{3^2+4^2}} = \frac{25}{5} = 5$
point-to-line distance / 點到直線距離

PROOF Distance = 5 = $r \Rightarrow L$ is a tangent to C **f.t.**
conclusion required / 必須寫結論

A1 $T = (6, 8)$
T = foot of perpendicular from G to L / T 為 G 至 L 的垂足

(c) (2 marks)

M1 $O = (0, 0)$ satisfies C ; mid-point of $OT = (3, 4) = G$ **f.t.** **alt: q19c_alt**
or show $OT = 10 = 2r$ / 或證 $OT = 10 = 2r$

A1 Mid-point of OT is the centre $G \Rightarrow$ **AGREE** **f.t.**
1A f.t. — conclusion required

(d) (2 marks)

M1 Taking OT as base, greatest height = $r = 5$ **f.t.**
height greatest when P is end of diameter perpendicular to OT

A1 $S_{\max} = \frac{1}{2}(10)(5) = 25$